Block Copolymer-Assisted Microcellular Supercritical CO_2 Foaming of Polymers and Blends

Michel Dumon[1,3], José Antonio Reglero Ruiz[1], Javier Pinto Sanz[2], Miguel Angel Rodriguez Perez[2], J-M. Tallon[3], M. Pedros[3], E. Cloutet[1], and P. Viot[4]

[1] *Laboratoire de Chimie des Polymères Organiques (LPCO), ENSCBP Ecole Nationale Supérieure de Chimie Biologie et Physique, 16 Avenue Pey-Berland, 33607 Pessac, Université de Bordeaux, France*
[2] *Cellular Materials Laboratory (CellMat), Condensed Matter Physics Dept, Universidad de Valladolid, Spain*
[3] *Département Science et Génie des Matériaux, IUT Institut Universitaire de Technologie, Gradignan, Université de Bordeaux, France*
[4] *Institut de Mécanique et d'Ingénierie de Bordeaux I2M, dépt DUMAS, Arts et Métiers, Talence, site de Bordeaux, France*

SUMMARY

The behaviour in supercritical CO_2 of block copolymers containing styrenic, butadiene, and methacrylic or perfluroalkyl blocks is studied in view of a specific swelling and foaming by a gas dissolution process. These block copolymers are considered as neat materials or as additives in blends e.g in polystyrene (PS) or polymethylmethacrylate (PMMA) matrices.

In both cases (neat or blend) the copolymers may exhibit a structuration at a micro or nano level. The phase separated (nano) structures depend on the block type and the concentration of copolymers in the polymer matrix, so that micelles, vesicles, lamellas, or warm-like structures are generated.

Furthermore when one block is chosen as a highly CO_2-philic moiety, the nanostructures are able to act as CO_2 reservoirs. The result is the possibility to control microcellular foaming, or sometimes nanocellular foaming, of commodity amorphous polymers such as PMMA and PS. Besides, at room temperature, the blocks can be either glassy or rubbery in order to freeze the growth and coalescence of cells during foaming.

Email addresses: michel.dumon@u-bordeaux1.fr, dumon@ipb.fr, jareglero@gmail.com, jpintosanz@gmail.com, marrod@fmc.uva.es, jean-marc.tallon@iut.u-bordeaux1.fr, matthieu.pedros@iut.u-bordeaux1.fr, cloutet@enscbp.fr, philippe.viot@ensam.eu

Different cellular polymers were elaborated by varying either the copolymer type or the foaming conditions (saturation pressure, temperature, depressurization rate). Cell sizes are accessible in a range from 0.2 to 200 μm, and densities from 0.40 to 1 g/cm³.

It is also shown that nanostructuring polymers are also efficient to produce polymer foams with oriented / structured voids.

This new approach could be used to produce nanocellular or ultra microcellular polymer foams in a simple process, using blending and extrusion.

INTRODUCTION

Microcellular polymeric foams are defined as porous materials having average cell sizes in the range of 1-10 μm, and cell densities on the order or larger than of 10^8 cells/cm^3. Nanocellular foams with cell sizes lower than 1 μm and cell densities between 10^9 and 10^{18} cells/cm^3 are now under development, bringing a real improvement in the mechanical and thermal behaviour of polymer foams, and maybe an expected transparency in the case of amorphous polymers. The syntheses are however non trivial; several techniques are known at the laboratory scale, among these techniques is the inert gas dissolution and consequent blowing, e.g. by CO_2. Particularly the low critical conditions of supercritical CO_2, (scCO_2 , 31.1°C and 73.8 bar) offer many advantages, like a tuneable solvent power, plasticization of glassy amorphous polymers and high diffusion rates [1,2]. The cellular microstructure, in any gas process, results from a balance between several phenomena:

i) Diffusion of CO_2 in the polymer (i.e. accessibility of CO_2).

ii) Affinity to CO_2 (i.e. solubility).

iii) Polymeric viscoelastic behaviour (deformations and viscosities) and the rubber-to-glass transition, T_g glass transition temperature (inducing vitrification on cooling, i.e. freezing in of the porous structures).

With the aim of elaborating nano porous polymers, a lot of papers have already shown the ability of various types of particles from submicron or nano size to act as nucleation sites and/or as foaming enhancers in either batch or continuous gas foaming processes: organo clays, silica in polystyrene, PMMA, … [3-5], or else elastomeric random copolymers or siloxane polymer additives or grafted PS / PMMA in polyolefins [6-9].

On the other hand, when polymer materials exhibit an organization at a macro, micro or nano level (e.g. semi crystalline polymers, liquid crystal polymers, (nano)structured polymers, polymer blends, …), the initial material properties may be beneficial either for the foam properties (create anisotropic foams) or for the foaming process (control of nucleation, stabilization of coalescence).

The case of block copolymers is more difficult because these polymers are able to self assemble, and also to induce a nanostructuration of a polymer matrix (by blending) [10-12]. Block copolymers are built with two or more polymer blocks of different chemical composition, length, and properties. The blocks are chemically linked to each other, preventing a macrophase separation. Self assembly / structuration / ordering of block copolymers result from the incompatibility of the blocks so that micro or nano phase separation takes place, which induces ordered periodic structures (spherical, micellar, lamellar, cylindrical, bicontinuous, …) depending on block composition, block length, block ratio [10-16].

Also the quality and the range of ordering is quite sensitive to process conditions (mass or solvent process, temperature annealing, extrusion shearing, type of surface for deposition) [13-16]. For example nanostructures are perfectly defined in solvent cast specimens and, although more poorly defined, the structures are still observed in bulk injected samples. The block MAM or SBM copolymers that will be used in our work are known to have a robustness for self-assembly in all process conditions [12,14,15,16].

The objective is to take advantage of the specific nanostructures of block copolymers (or blends of block copolymers with an homopolymer) to change both the gas foaming behaviour (nucleation –homogeneous vs. heterogeneous, coalescence, gas uptake and gas localization, selective vitrification of the nanostructure) and the final foam structures (cell size, cell density, cell homogeneity and isotropy), especially towards new nano foams [17-20].

In the field of micro or nano porous organic materials, so far only *nanoporous organic thin films* were mainly produced [21-23]. In another field (inorganic porous materials), organic block copolymer templating is well known in inorganic or hybrid porous materials.

Previous works dealing with micro cellular *bulk* polymer foams are indeed not numerous [24-26].

Following this line of investigation, the use of supercritical carbon dioxide as a physical foaming agent is investigated in several CO_2-philic nanostructuring block copolymers, both neat or after blending with homo polymers (the homopolymers are amorphous commodity polymers, PS, PMMA) [27,24].

A rather easy process of bulk cellular materials is carried out in a batch single step-process. In this process, the polymer is saturated with CO_2 in the supercritical regime, during a fixed time at a chosen temperature. After saturation, the sample, approaching or being in a rubbery state, is depressurized to atmospheric pressure, taking advantage of the swelling and plasticization of the polymer, which reduces the glass transition temperature, allowing the gas expansion. In this method, the cellular structure is controlled by changing the saturation temperature and/or the depressurization rate, but moreover by selecting the type of CO_2-philic block in the copolymer, especially its affinity for CO_2 [29-33].

EXPERIMENTAL

Materials

Commercial block acrylate copolymers (SBM; MAM, SM), one lab-synthesized fluoro acrylate copolymer and two neat homopolymers were used. A transparent polymethylmethacrylate (PMMA, M_w=70000 g/mol) was kindly supplied by Arkema, in the form of pellets, together with a high molar mass transparent polystyrene (PS, M_w=140000 g/mol), which was kindly supplied by Total Petrochemicals company, in the form of pellets. Commercial block copolymers were supplied by Arkema Company, in the form of pellets, and employed as received [12, 14-16].

- A triblock terpolymer, *SBM* (polystyrene-b-polybutadiene-b-polymethylmethacrylate named *SBM*), with a composition of 52 wt% stryrene, 30 wt% butadiene, and 18 wt% methylmethacrylate with a PS block of molar mass equal to 30000 g/mol.
- A triblock terpolymer, *MAM* (polymethylmethacrylate-b-polybutylacrylate-b-polymethylmethacrylate named *MAM*) with 35 wt% methacrylate, 30 wt% butylacrylate and 35%wt methacrylate, and a molar mass equal to 90000 g/mol.
- A di block copolymer, *SM* (polystyrene-b-polymethylmethacrylate named *SM*) with a composition of 58 wt% stryrene and 42 wt% methylmethacrylate with a molar mass equal to 90000 g/mol.
- A di block fluoro copolymer, *PS-b-PFDA*. It is well known that fluorinated polymers present the highest CO_2 – phicility [21-23] and in the last years, several efforts have been carried out to obtain nanocellular porous films through CO_2 philic fuoro blocks. This route is valuable if a rather easy synthesis is used. ATRP, Atom Transfer Radical Polymerization is suggested [28] to synthesize diblock copolymers formed by a polystyrene (PS) block and fluorinated blocks (e.g. a perfluorodecyl acrylate block, *PFDA*).

- Blends were produced by extrusion : PS + SBM; PMMA + MAM.

In all cases, blending was carried out by extrusion followed by injection moulding to obtain bulk specimens in the form of parts of 15x60 mm^2 with a 1.5 mm thickness.

Blends of PS+MAM and PMMA+SBM were also produced, but due to the complete immiscibility with the homopolymer, samples were phase-separated at a macro-scale level, showing no transparency with a heterogeneous foam structure. For this reason, these materials were rejected for micro foaming purposes.

Foaming Procedure and Structure Observations

Microcellular foaming experiments were carried out in a high pressure reactor provided by TOP Industry (France), with a capacity of 300 cm^3 and capable of operating at 250°C and 400 bar. The reactor is equipped with a pressure pump controller provided by Teledyne ISCO, and controlled to maintain automatically the temperature and pressure at a given values. The CO_2 vessel temperature and pressure were monitored in the course of the whole process. The samples were saturated with supercritical CO_2 at a maintained pressure of 300 bar during 24 h. Saturation temperatures were varied between 25°C and 80°C (and one specific experiment at 0°C in an ice bath). It is important to notice that the weight of gas absorbed between 16 h and 24 h remains invariable in our samples (1.5 mm thickness).

Finally because CO_2 absorption increases with decreasing temperature, we also intended to incorporate CO_2 at 0°C (PS-PFDA) in order to initiate foaming in the fluoro nuclei surrounded by a rigid glassy poorly CO_2-philic matrix (e.g. PS) to imitate the growth stage.

After the saturation process, foaming was triggered by depressurization at a chosen constant rate (between 10 to 150 bar/min). Samples were removed from the vessel when reaching room temperature and kept 30 min before preparation for SEM observations.

Cellular structure was characterized by means of Scanning Electron Microscopy (model HITACHI S-3000N), on liquid nitrogen fractured samples. Dense solid nanostructures were observed by Transmission Electron Microscopy TEM at the Bordeaux Imaging Center (Université Bordeaux 2 Segalen) on ultra microtomed thin films (≈80 nm), treated either by RuO_4 or phosphotungstic acid. The help of Dr. Etienne Gontier and Dr. Soizic Lacampagne is indeed acknowledged.

RESULTS

Structures of copolymers and blends, Absorption of CO_2

Nanostructures

The nano structures of bulk injected specimens (15x60x1.5 mm) are examined in the direction vertical (V) to the flux or horizontal (H). **Figures 1, 2, 3** give TEM micrographs of blends of PMMA and MAM copolymer showing that nanostructures are retained in any of the compositions. The neat MAM copolymer has a lamellar structure; whereas the 50/50 blend exhibit worm like structures, even showing individual blocks. The 90/10 and 95/5 blends show micellar structures where the polybutylacrylate units are within the micelles surrounded by PMMA shells. Those structures are oriented by the flow during injection moulding.

Therefore CO_2-philic blocks are individualized both in neat polymers or in blends and might act as CO_2 concentrators and nuclei.

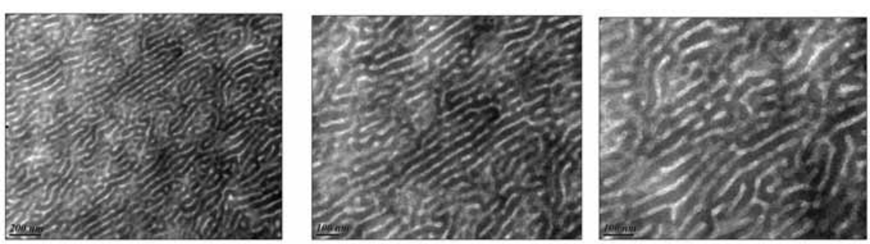

***Figure 1**. TEM micrographs of neat MAM copolymer (vertical cut of injected specimen), bar scale 200 or 100 nm*

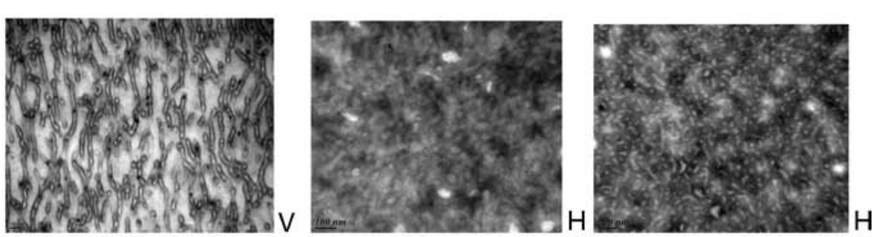

***Figure 2**. TEM micrographs of PMMA / MAM blends (50/50%wt), bar scale 100 or 200 nm. Left micrograph: (V) vertical cut of injected specimen, right micrographs: (H) horizontal cut of injected specimen*

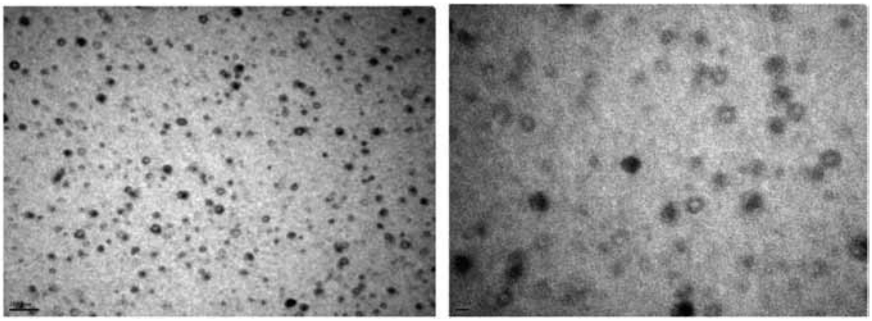

***Figure 3**. TEM micrographs of PMMA / MAM blends (95/5%wt) (horizontal cut of injected specimen), bar scale 100 and 20 nm*

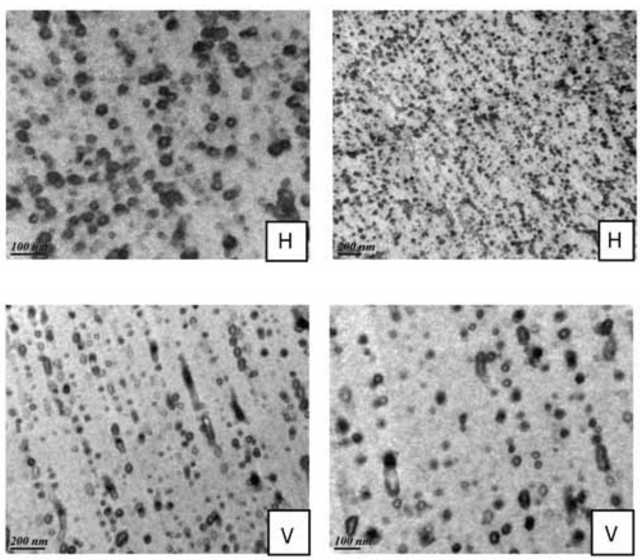

***Figure 4**. TEM micrographs of PMMA / MAM blend (90/10%wt), bar scale 100 and 200 nm. Top micrographs: (H) horizontal cut of specimens, bottom micrographs: (V) vertical cut of specimens*

CO_2 Absorption

The CO_2 mass uptake was measured experimentally according to time or saturation temperatures (**Figure 5**, **Table 1**). An example of mass uptake is given on **Figure 5** where experimental mass data were obtained after removing the samples from the CO_2 vessel and immediately transferring them to a high precision balance, carrying out several measurements at different times to evaluate the gas loss out of the vessel due to diffusion process. The measured values are an estimation of the maximum quantity of gas uptake.

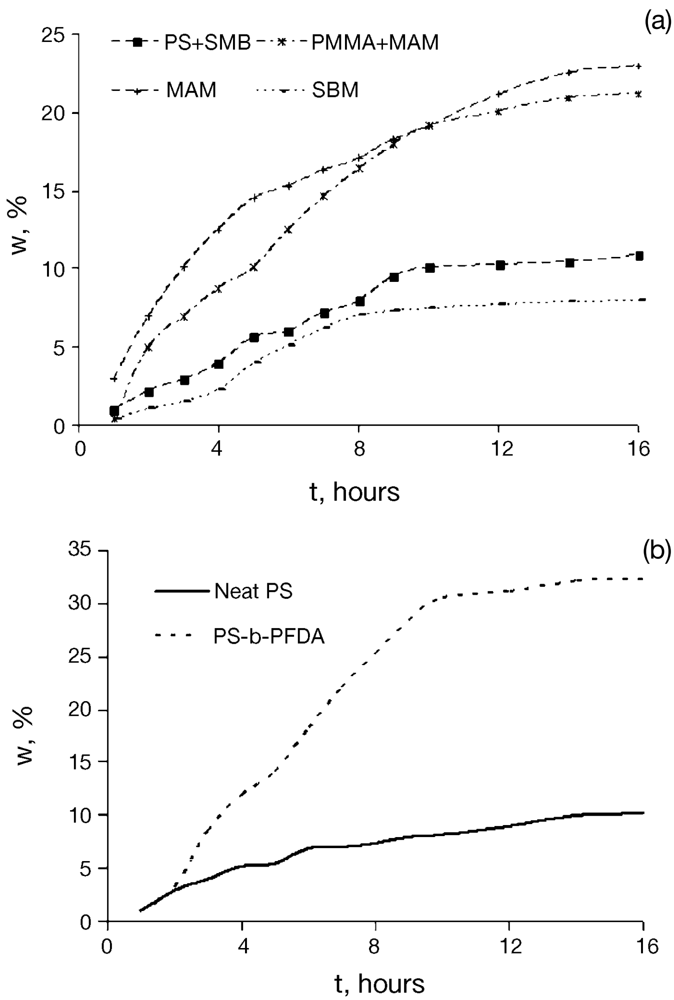

***Figure 5**. CO_2 mass uptakes measured at RT, out of the vessel, after $scCO_2$ exposure at different times (saturation values, 300 bar). a) $T_{saturation}$=40°C for copolymers and blends. Styrene base: upper curves, Acrylate base: lower curves. b) $T_{saturation}$=0°C for a fluoro PS-b-PFDA copolymer (50 wt% PFDA)*

The quantity of CO_2 uptake, *w,* is quantifying the affinity of CO_2 for the material considered; this quantity plays a role on the final foam density and on the plasticization of the polymer, lowering the T_g of the polymer due to CO_2 swelling, which in turn delays the vitrification of the polymer during foaming, thus allowing more time for cell growth. Fluoro based polymers, i.e. PS-PFDA, is the most CO_2-philic polymers in the list chosen and PMMA and PMMA-based copolymers are, as expected, among the best CO_2-philic systems compared to neat PS or polyolefins.

Table 1. Weight gain ratio of CO2 after saturation at different temperatures (saturation conditions 300 bar, 16 h)

Material	$w_{25°C}$ (%)	$w_{40°C}$ (%)	$w_{50°C}$ (%)	$w_{60°C}$ (%)	$w_{70°C}$ (%)	$w_{80°C}$ (%)
PS	10.4	10.3	10.1	9.4	9.2	8.9
PMMA	16.9	16.4	16.2	16.1	15.9	15.6
PS+SBM	12.1	10.9	10.5	10.4	10.2	9.7
PMMA+MAM	22.3	21.2	20.4	20.1	19.3	19.2
SM	12.5	12.3	12.0	11.2	11.1	10.8
MAM	24.1	22.9	22.4	22.4	22.1	21.9
SBM	8.3	8.1	8.0	7.6	7.6	7.4
PS-b-PFDA	30					

Microcellular Foaming Process

When the two different foaming routes are analysed, they are shown to have quite different vitrification capabilities. In the first route, for a fixed saturation temperature, **Figure 6a**, cells are formed by triggering the depressurization drop rate between 150 bar/min to 12 bar/min, thus inducing a pressure drop ΔP, from the same initial quantity of swollen CO_2, in samples of exactly the same geometry. In the second route, **Figure 6b**, depressurization rate is fixed at 60 bar/min, whereas cells are formed by pressure release from a fixed saturation temperature which was changed between 25°C to 80°C. The two routes do show very different temperature drops. The extent of this temperature drop depends on factors such as reactor volume, thermal conductivities, sample mass or pressure drop rate. Nevertheless, for the "ΔP

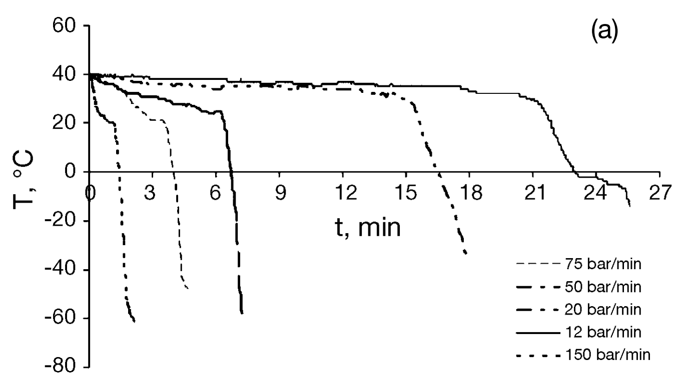

***Figure 6**. Typical temperature drop curves in the one-step CO_2 process. a) Saturation temperature 40°C, variable depressurization rate*

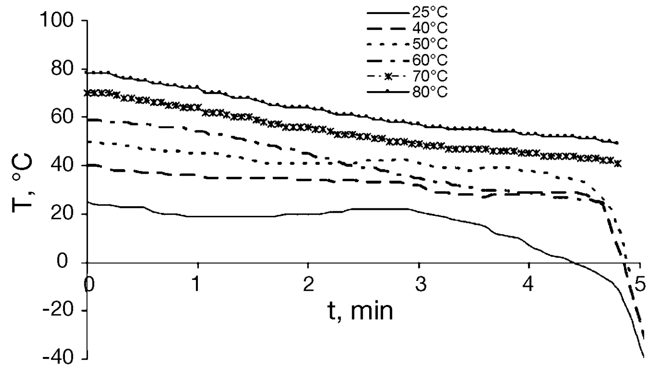

Figure 6. Typical temperature drop curves in the one-step CO_2 process. b) Fixed depressurization rate of 60 bar/min, saturation temperature from 25°C to 80°C

route", the curve exhibits a sudden temperature drop probably corresponding to the supercritical - liquid transition (**Figure 6a**).

It is revealed that temperature inside the reactor may be decreased greatly: down to -50°C at the end of the process (for a depressurization rate of 150 bar/min), and down to -10°C for the lowest depressurization rate (12 bar/min). Therefore, the onset of vitrification is modified and arrives much earlier if ΔP is increased. Thus the cells are frozen in at different times.

Table 2 presents average cell characteristics for both PS and PMMA based foams. Foaming of PS is only possible above 50°C, and average cell sizes are

Table 2. Foam characteristics of PS or PMMA based polymers

	PS + SBM 10%			PMMA + 10%MAM		
$\Delta P/\Delta t$ bar/min	Φ (μm)	ρ_{foam} (g/cm^3)	N_c (cells/cm^3)	Φ (μm)	ρ_{foam} (g/cm^3)	N_c (cells/cm^3)
150				0,3	0,71	2,1 10^{13}
75				2	0,65	8,6 10^{10}
50				10	0,63	7,3 10^{8}
20	12	0,90	1,2 10^{8}	20	0,60	9,8 10^{7}
12	20	0,82	4,5 10^{7}	80	0,53	1,7 10^{6}
T_{sat} (°C)						
25				1	1	3,7 10^{10}
40				2	0,93	4,8 10^{10}
50				3	0,92	1,5 10^{10}
60	15	0,94	3,9 10^{7}	4	0,83	5,3 10^{9}
70	20	0,85	3,8 10^{7}	5	0,78	3,5 10^{9}
80	30	0,76	1,8 10^{7}	7	0,65	2 10^{9}

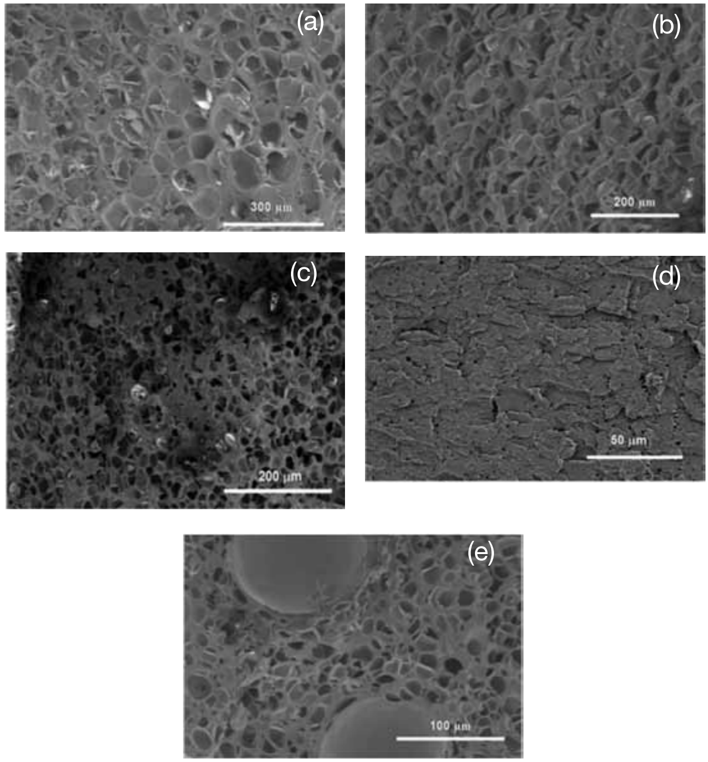

***Figure 7**. SEM micrographs of the microcellular foams obtained for neat copolymers (T_{sat} = RT or T_{sat} = 70°C). a) MAM at 70°C, b) MAM at RT, c) SBM at 70°C, d) SBM at RT, e) SM at 70°C*

ranging between 10 µm and 30 µm. For PMMA, foaming is possible all over the temperature range, with a wider cell size range, but down to 0.3 µm. This is again related to the higher CO_2 affinity of PMMA compared to PS, which allows the production of microcellular foams all over the temperature range.

Block Copolymer Role on Foaming

Indeed, even if the SBM copolymer is dispersed at a nanometer level, it does not enhance greatly the CO_2 absorption and SBM does not improve the cellular structure at low foaming temperatures. However, increasing the process temperature up to 80°C results in a very homogeneous cellular structure, with a lower average cell size (around 30 µm) and a higher number of nucleated sites than in neat PS.

Figure 8 presents several SEM micrographs showing the cellular structure of the PMMA + MAM (90/10 wt%) foams produced at 25°C and 80°C. MAM enables both a higher CO_2-philicity than PMMA and a rubbery butyl acrylate block. Thus a completely different foaming behaviour is obtained.

Figure 9 shows the variation of average cell size ϕ with saturation temperature for PS+SBM or PMMA+MAM foams.

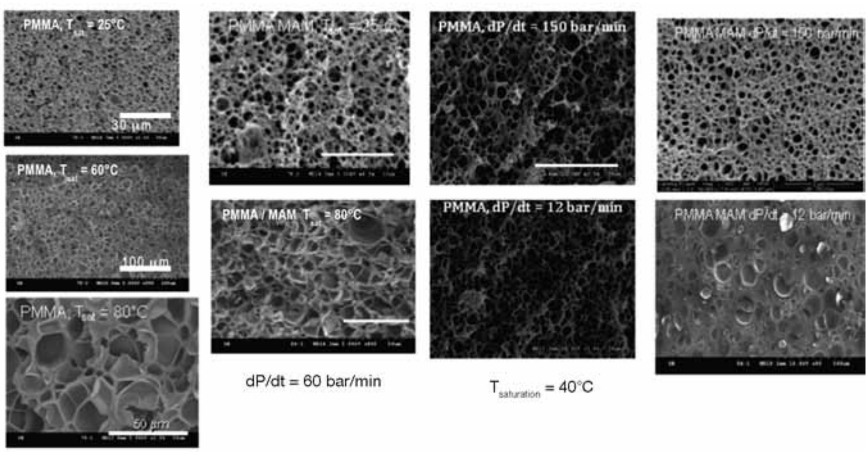

Figure 8. SEM micrographs of foams obtained for PMMA or PMMA-MAM 90/10 wt% blends (T_{sat} = 40°C)

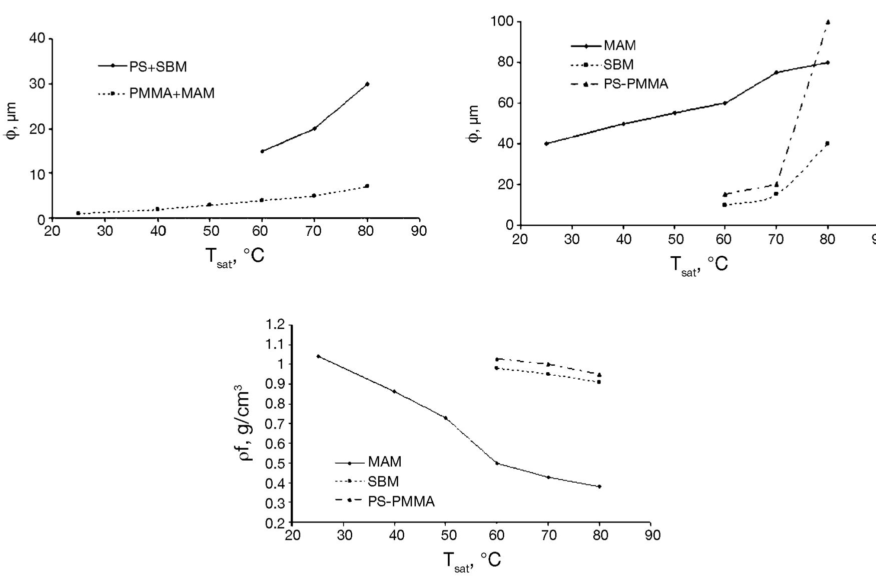

Figure 9. Average cell size or density in copolymer and blend foams

Finally two interesting points are still under study:

- A direct relation between the initial structure, for example in a PS-b-PFDA copolymer and the resulting foam as shown in **Figure 10**.
- Unconventional foams (e.g. lamellar) may be generated in certain conditions (at low temperature 40°C, for contents of MAM > 50 wt%) as shown in **Figure 11**.

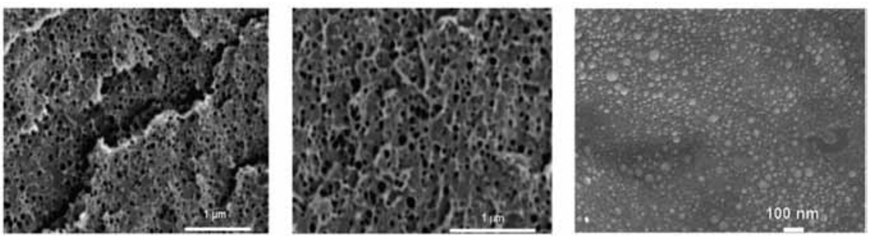

Figure 10. *SEM micrographs of the microcellular foams obtained for a PS-PFDA copolymer (T_{sat} = 40°C)*

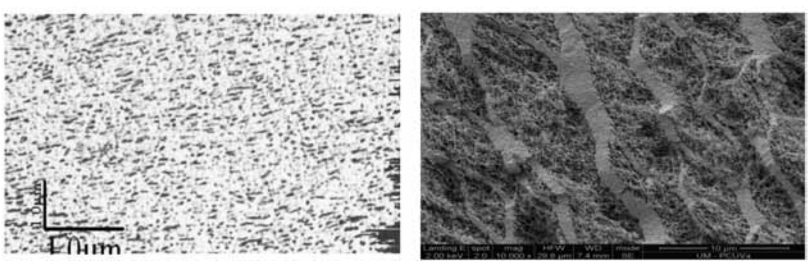

Figure 11. *PMMA/MAM blend (25/75) : before foaming (left,–AFM map); after foaming (right,–SEM micrograph)*

CONCLUSIONS

Microcellular foaming of amorphous polymers, e.g. Polymethylmethacrylate and Polystyrene, was studied in supercritical CO_2 via a single-step in the presence of several types of block copolymer additives able to change their foaming behaviour and therefore the porous structures. Some of them are able to enhance greatly the CO_2 uptake (rubbery acrylate, fluoro copolymers) and change the foaming behaviour, modifying the cell size. The observed behaviours are explained in terms of interplay between CO_2-philicity and vitrification of the polymer matrix or vitrification of one poorly CO_2-philic block, together with the resulting plasticization delay, i.e. the period or temperature range (ΔT) in which the polymer gas system remains rubbery.

Even in the presence of well dispersed but poorly CO_2 compatible additive (e.g. SBM), an improvement in the cell size reduction and foam homogeneity is observed. If the additive is highly CO_2-philic, as a MAM copolymer, a drastic reduction of the average cell size is observed, producing ultra-microcellular foams. In this latter case, the CO_2-philic dispersed entities (MAM block copolymer) having an always rubbery butyl acrylate block provides a good compromise between CO_2 phicility, nano sized dispersion of the additive (role of nuclei), and possibly a cell growth limitation within the additive (due to surrounding by the other glassy PMMA blocks).

REFERENCES

1. Production of controlled polymeric foams by supercritical CO_2, E. Reverchon and S. Cardea, *J. of Supercritical Fluids*, **40**, 144–152 (2007).

2. Change of glass transition temperature of polymers containing gas, J. D. Yoon, S. W. Cha. *Polymer Testing*, **20**(3), 287-29 (2001).

3. Layered-silicate based polystyrene nanocomposite microcellular foam using supercritical carbon dioxide as blowing agent, Bin Zhu, Weibin Zha, Jintao Yang, Cailiang Zhang, L. James Lee, *Polymer* **51,** 2177-2184 (2010).

4. Synthesis and processing of PMMA carbon nanotube nanocomposite foams, Changchun Zeng, Nemat Hossieny, Chuck Zhang, Ben Wang, *Polymer* **51,** 655–664 (2010).

5. Controlled Foaming of Polymer Films through Restricted Surface Diffusion and the Addition of Nanosilica Particles or CO_2-philic Surfactants, Srinivas Siripurapu, Joseph M. DeSimone, Saad A. Khan, Richard J. Spontak**,** *Macromolecules***, 38***,* 2271-2280 (2005).

6. The effect of dispersed elastomer particle size on heterogeneous nucleation of TPO with N_2 foaming, Seong G.Kim, Siu N.Leung, Chul B Park, Mohini Sain, *Chemical Engineering Science*, **66**, 3675–3686 (2011).

7. Control of bubble size and location in nano/micro scale cellular PP/rubber blend foams, T Nemoto, J Tagaki, M Ohshima, *Macrom. Mol. Eng. Sci.* **293**, 574-580 (2008).

8. Effect of the Introduction of Polydimethylsiloxane on the Foaming Behavior of Block-Copolymerized Polypropylene, Mingyi Wang, Jun Ma, Raymond Chu, Chul B. Park, Zhou Nanqiao, *Journal of Applied Polymer Science*, **123**, 2726–2732 (2012).

9. Foaming behavior of isotactic polypropylene in supercritical CO2 influenced by phase morphology via chain grafting, Wentao Zhai, Hongying Wang, Jian Yu, Jin-Yong Dong, Jiasong He, *Polymer* **49,** 3146–3156 (2008).

10. Directing the self-assembly of block copolymers, S.B. Darling, *Prog. Polym. Sci.* **32,** 1152–1204 (2007).

11. Controlling the size of nanostructures in thin films via blending of block copolymers and homopolymers, Juan Peng, Xiang Gao, Yuhan Wei, Hanfu Wang, Binyao Li, and Yanchun Han, *The J. of Chem Phys.* **122**, 114706(1-7) (2005).

12. Molecular Disorder and Mesoscopic Order in Polydisperse Acrylic Block Copolymers Prepared by Controlled Radical Polymerization, A.V Ruzette, S. Tencé-Girault, L. Leibler, F. Chauvin, D. Bertin, O. Guerret, P. Gérard, *Macromolecules*, **39**, 5804-5814 (2006).

13. Morphology and mechanical properties of poly(methylmethacrylate)-bpoly(alkylacrylate)-b-poly(methylmethacrylate), J.D. Tong, Ph. Leclere, C. Doneux, J.L. Bredas, R. Lazzaroni, R. Jérome, *Polymer* **42,** 3503-3514 (2001).

14. Microdeformation mechanisms in rubber toughened PMMA and PMMA-based copolymers, Laure Lalande, Christopher J.G. Plummer, Jan-Anders E. Manson, Pierre Gerard, *Engineering Fracture Mechanics* **73**, 2413–2426 (2006).

15. Controlled Architecture Polymers at Arkema: Synthesis, Morphology and Properties of All-Acrylic Block Copolymers, P. Gerard, N. P Boupat, T. Fine, L. Gervat, J-P. Pascault, *Macromol. Symposium* 256, 55-64 (2007).

16. Annealing and Defect Trapping in Lamellar Phases of Triblock Terpolymers, Laurent Corte , Kazuhiro Yamauchi, Francois Court, Michel Cloıtre, Takeji Hashimoto, Ludwik Leibler, *Macromolecules***, 36***,* 7695-7706 (2003).

17. Investigation of a Strategy for Well Controlled Inducement of Microcellular and Nanocellular Morphologies in Polymers, Hamed Janani, M.H.N. Famili, *Polym Eng and Sci* **10**, 1558-1570 (2010).

18. Block Copolymers as Precursors to Porous Materials, T. P. Russell and J. L. Hedrick, *Encyclopedia of Materials: Science and Technology* ISBN: 0-08-0431526, pp. 692-695 (2001).

19. Block Copolymer Micelles for Nucleation of Microcellular Thermoplastic Foams, Pieter Spitael, Christopher W. Macosko, Richard B. McClurg, *Macromolecules* ***37,*** 6874-6882 (2004).

20. Thermodynamics and Kinetic Processes of Polymer Blends and Block Copolymers in the Presence of Pressurized Carbon Dioxide, Teri A. Walker, David J. Frankowski, Richard J. Spontak, *Adv. Mater.* **20**, 879–898 (2008).

21. Thermally robust nanocellular thin films of high-Tg semifluorinated block copolymers foamed with supercritical carbon dioxide, Rui Zhang, Cedric Dutriez, Kenji Sugiyama, Takashi Ishizone, Hideaki Yokoyama, *Soft Matter*, **7**, 4032–4038 (2011).

22. Nanocellular structures in block copolymers with CO_2-philic blocks using CO_2 as a blowing agent: Crossover from micro- to nano-cellular structures with depressurization temperature, H. Yokoyama, K. Sugiyama, *Macromolecules* **38**, 10516-10522 (2005).

23. Nanocellular Foams of PS/PMMA Polymer Blends, T. Otsuka, K. Taki, M. Ohshima, ***Macromol. Mater. Eng.*** 293, 78-82 (2008).

24. Foaming behaviour and compressive properties of microcellular nanostructured polystyrene, J.A. Reglero, P. Viot, M. Dumon, ***Cellular Polymers,*** **28(6)**, 363-386 (2009).

25. Micro and nano cellular amorphous polymers (PMMA, PS) in supercritical CO_2 assisted by nanostructured CO2-philic block copolymers – One step foaming process, J.A. Reglero, J.M. Tallon, M. Pedros, M. Dumon, ***Journal of Supercritical Fluids,*** **58**, 168-176 (2011).

26. Low-Density Nanocellular Foams Produced by High-Pressure Carbon Dioxide, J.A. Reglero, M. Dumon, J. Pinto and M.A Rodríguez-Perez, ***Macromolecular Materials and Engineering,*** **296**, 752-759 (2011).

27. Microcellular foaming of polymethylmethacrylate in a batch supercritical CO_2 process: Effect of microstructure on compression behavior, J.A. Reglero Ruiz, P. Viot, M. Dumon, ***Journal of Applied Polymer Science,*** **118**(1), 320–331 (2010).

28. "Investigation of the Nanocellular Foaming of Polystyrene in Supercritical CO_2 by Adding a CO_2-Philic Perfluorinated Block Copolymer", J.A. Reglero Ruiz, E. Cloutet, M. Dumon, Journal of Applied Polymer Science, **126**, 38–45 (2012).

29. In Situ IR Spectroscopy and Ab Initio Calculations To Study Polymer Swelling by Supercritical CO_2, Pauline Vitoux, Thierry Tassaing, Francois Cansell, Samuel Marre and Cyril Aymonier, ***J. Phys. Chem. B*, *113,*** 897–905 (2009).

30. Solubility and Self-Assembly of Amphiphilic Gradient and Block Copolymers in Supercritical CO_2, Tiphaine Ribaut, Julian Oberdisse, Burkhardt Annighofer, Bruno Fournel, Stephane Sarrade, Heike Haller, Patrick Lacroix-Desmazes, ***J. Phys. Chem. B*** **115**, 836–843 (2011).

31. Solubility of fluorinated homopolymer and block copolymer in compressed CO_2, Pascal André, Patrick Lacroix-Desmazes, Darlene K. Taylor, Bernard Boutevin, ***J. of Supercritical Fluids*** **37**, 263–270 (2006)

32. Foaming of amorphous polymers and blends in supercritical CO_2: Solubility versus block copolymers addition, Jose Antonio Reglero, Philippe Viot and Michel Dumon, ***Journal of Cellular Plastics*** **47,** 535-548 (2011).

33. Two-step micro cellular foaming of amorphous polymers in supercritical CO_2, Jose Antonio Reglero Ruiz, Jean Marc-Tallon, Matthieu Pedros, Michel Dumon, ***J. of Supercritical Fluids*** **57**, 87–94 (2011).